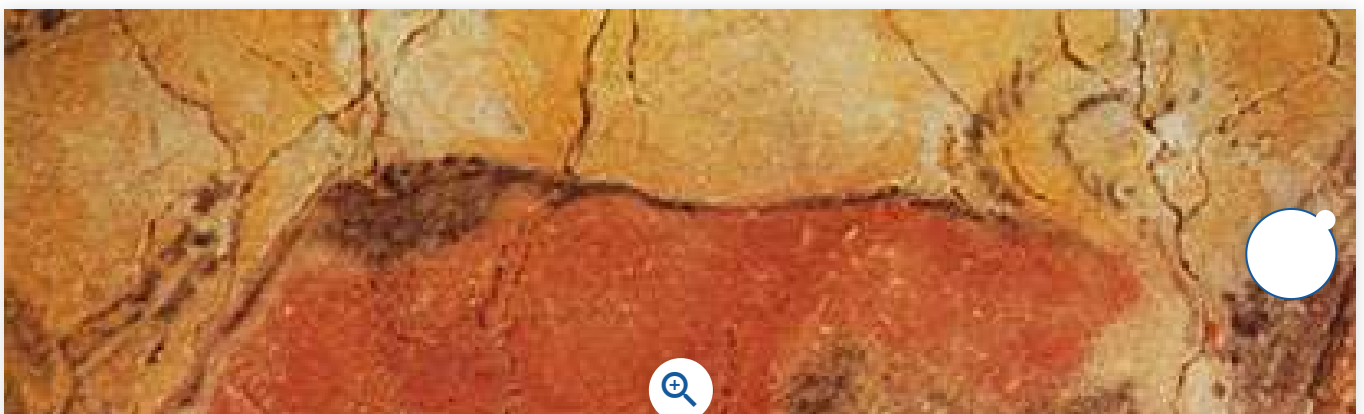


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Language, culture, and lifeways in the Pleistocene

Speech and symbolic intelligence

The origin and development of [human](#) culture—articulate spoken [language](#) and symbolically mediated ideas, beliefs, and behaviour—are among the greatest unsolved puzzles in the study of human [evolution](#). Such questions cannot be resolved by skeletal or archaeological data. Research on the [behaviour](#) and [cognitive](#) capabilities of [apes](#), [monkeys](#), and other [animals](#) and on cognitive development in human children provide some clues, but [extrapolating](#) this information back through time is [tenuous](#) at best. Complicating the scenario further, it may be that today's [chimpanzees](#), [bonobos](#), and other anthropoid [primates](#) have more sophisticated cognitive capabilities and behavioral skills than those of some early [hominins](#), because they and their ancestors have had several million years to overcome many challenges and perhaps have become more advanced in the process. [Speech](#) has been inferred by some investigators on the basis of certain internal [skull](#) features, for example, in *H. habilis*, but [jaw](#) shape and additional [traits](#) suggest otherwise. Still other researchers claim that human speech was not even fully developed in early members of anatomically modern *H. sapiens*, because of the simplicity of their [tool](#) kits and [art](#) before the [Late Paleolithic](#).





Magdalenian cave painting of a bison

Magdalenian cave painting of a bison, Altamira, Spain.

Image: A. Held/J.P. Ziolo, Paris

It is impossible to assess linguistic competency by observing the insides of reassembled fossil craniums that are incomplete, battered, and distorted—and in any case the brains probably did not fit snugly against the walls of the braincase. The apparent cerebral expansion in *H. habilis* and *H. rudolfensis* may imply a general increase in cognitive abilities, manipulative skill, or other factors besides speech. Particularly unreliable are claims that the specific internal cranial impressions of a Broca cap is evidence of speech. Prominent Broca caps exist among some chimpanzees, yet no ape has uttered a word, despite laborious attempts to get them to speak.

A humanoid vocal tract is undetectable in fossils because it comprises only soft tissues and leaves no bony landmarks. Although versatile human speech is reasonably linked to a relatively spacious pharynx and mobile tongue, the absence of such features is not a compelling reason to deny some form of vocal language in ancestral hominins. It is argued that articulate human speech is impossible without a lowered voice box (larynx) and an expanded region above it. If this presumption were true, even Neanderthals would be inept vocally and probably also quite primitive cognitively as compared with Late Paleolithic *H. sapiens* populations such as the Cro-Magnons. Gibbons and great apes do not speak, yet they have throat traits concomitant with speech, albeit to a lesser degree than humans'. The calls of gibbons are wonderfully varied in pitch and pattern, and, if such sounds were broken into discrete bits with consonants, they could emulate words. The same may be said of great apes. Orangutans, chimpanzees, and bonobos have sufficiently mobile lips and tongues; they simply lack neural circuitry for speech.



Cro-Magnon

Artist's reconstruction of a Cro-Magnon, an early modern human in Europe.

Image: Courtesy of the American Museum of Natural History, New York

Conversely, if the theory that different abilities are governed by distinct and separate forms of intelligence ([multiple intelligences](#)) is correct, much of tool-using behavior and artistic ability would have to be based upon neurological structures fundamentally different from those that support verbal ability. Human children

begin to use language before they become sophisticated tool users. Similarly, a form of speech might have preceded forms of tool behaviour that are symbolically mediated. Visual arts such as [painting](#) and [sculpture](#) are expressions of spatial intelligence, which is centred principally in areas of the brain different from those related to speech. Therefore, one cannot expect the problem of language origins or language competence to be clarified by studying Paleolithic [symbolism](#) and imagery, despite the awesome array of [cave art](#) and polished [bone](#), antler, [ivory](#), [stone](#), and shell [artifacts](#) associated with the period. Yet if the stunning proliferation and stylistic variability of tools, bodily ornaments, and artistic works during the Paleolithic do not point unequivocally to the specific use of speech, the presence of these symbolically mediated artifacts—among the earliest of which are shell beads found in [Morocco](#) and made about 82,000 years ago—does indicate that early humans were capable of complex [conceptual](#) and abstract [thought](#).

Historically, all human groups [manifest](#) rich symbolically mediated language, [religion](#), and social, political, and [economic systems](#), even in the absence of elaborate [material culture](#). The demands on the social intelligence of peoples who live in [environments](#) with relatively few artifacts are similar to demands placed upon those who depend upon complex technological gadgets and shelters for comfort. Consequently, prehistoric *H. sapiens* cannot be regarded as cognitively less capable than ourselves, and it is impossible to state which [hominin](#) species were “fully human” as [symbol](#) users. As a case in point, meticulously documented language studies of captive bonobos and chimpanzees demonstrate that they have the capability to comprehend and use symbols in order to communicate with humans and with one another, but the use of this potential in the wild remains to be demonstrated. Perhaps the human capacity symbolically to represent [feelings](#), situations, objects, and [ideas](#) developed before being commandeered by the several intelligences and before it became a boon to vocal communication.

Archaeological evidence indicates that, like at least some of their [Pliocene](#) predecessors, the most recent [hominins](#) were probably [omnivorous](#), though how much [meat](#) was in their diets and whether they obtained it by scavenging, [hunting](#), or both are poorly documented until about 200–100 kya. Stone [tools](#) and cut marks on

[bones](#) at archaeological sites attest to a long history of meat eating in the tribe

Hominini, but this practice could have existed long before stone tools were invented. Like [chimpanzees](#), [bonobos](#), [baboons](#), [capuchins](#), and other [primates](#), early Pliocene hominins may have killed and fragmented [vertebrate](#) prey with only their [hands](#) and [jaws](#) instead of tools. The extent to which our ancestors' hunting, scavenging, or other activities were communal and coordinated via symbolic communication has not been determined.

There is no valid way to estimate group size and [composition](#) because there is little evidence of movement patterns, shelters, and graves until the [Late Paleolithic](#). Archaeological traces of human-made shelters occur rarely from 60 kya, then become more common, particularly in regions with notable seasons of inclement weather. The first appearances and development of symbolically based spirituality are also highly [elusive](#) because they left no morphological or unarguable archaeological trace until the [innovation](#) of writing and ritual paraphernalia; however, there is evidence that Neanderthals used jewelry and other personal ornaments some 44,000 years ago. Although some [Neanderthals](#) buried their dead, there is little evidence of mortuary ceremony in their graves. Graves of *H. sapiens* from 40 kya sometimes contain grave goods.

Learning from the apes

[Gorillas](#), [chimpanzees](#), and bonobos are a rich resource for cultural anthropologists, biologists, and psychologists who speculate on the origins of human society. Gorillas appeal to theorists who stress male dominance and [patriarchy](#). A characteristic [gorilla](#) group has one [silverback](#) (an older dominant male), one or more subordinate blackback males, [adult](#) females outnumbering males, and youngsters of various ages. The silverback is the hub of the [cohesive](#) group. Chimpanzee society is also dominated by males, which form a stable core of the group. Chimpanzees and bonobos live in larger groups numbering more than 100 individuals, though they forage, travel, and nest in much smaller bands that vary daily in number and composition. Among chimpanzees there is a top male, followed by several others whose ranks depend upon which other males are present. Bonobos have stronger

amulations between males and females than chimpanzees do, and the organizational

hub of [bonobo](#) social groups is based on [intimate](#) relations among adult females, particularly mothers, which often retain strong bonds with their sons. Adult male bonobos are less strongly bonded with one another than [chimpanzee](#) males are. Because bonobos are more pacific and tolerant in social relationships and are highly sexual, they are popular with those who would model our heritage as free of “killer apes.” However, observers of apes, Old World [monkeys](#), and other [mammals](#) have documented incidents of [aggression](#) as well as concern for others in their subjects. Both tendencies are deeply rooted among the higher primates.

The [emergence](#) of the human [nuclear family](#) has been a particularly knotty problem for Western evolutionary theorists. Like bonobos and chimpanzees, people probably are fundamentally promiscuous, though such [mating](#) behaviour is heavily proscribed by the [cultures](#) into which individuals are born and reside. Indeed, theorists who wish to construct models of the emergence of hominin societies on the basis of [extant](#) [ape](#) societies seldom tackle the overriding fact that humans utilize a wide variety of [kinship](#), social, sexual, and political arrangements, all of which are maintained and expressed symbolically as well as practically. Researchers often fail to search for the cognitive basis of symbolic representation, manipulation, and invention in apes, citing instead forms of behaviour that appear to [harbinger](#) specific human conditions. It will take the efforts of several scientific [disciplines](#) and sophisticated [technology](#), probably over many years, to discover the underlying nature of our mental faculties, their neurological basis, and their development over time. Apes can play important roles in this enterprise only if they are allowed to survive in their natural habitats and only if they are viewed as being on their own evolutionary paths and not merely as steps toward the human condition.

Russell Howard Tuttle

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